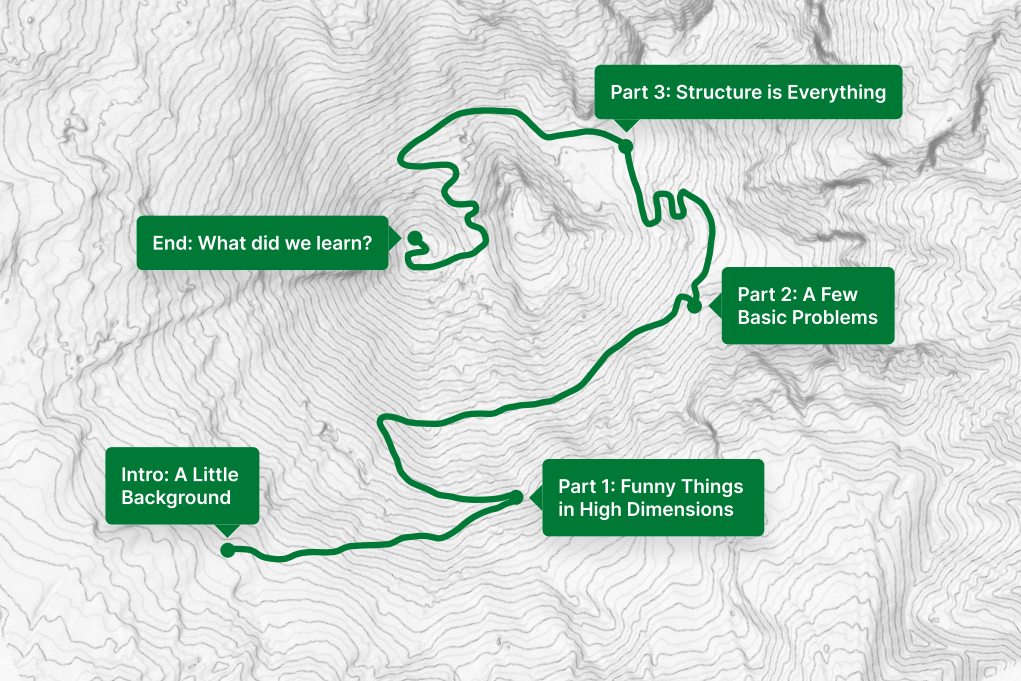
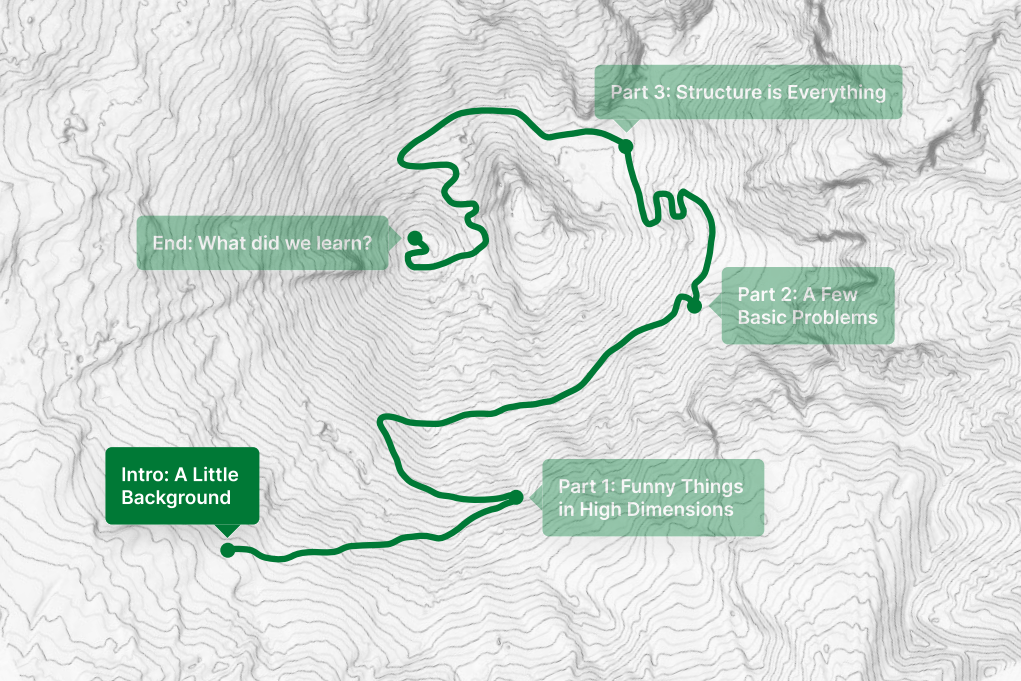
A Short Hike Through High-Dimensional Statistics

Giovanni Ballarin

DSF 2026




Part 3: Structure is Everything
End: What did we learn?
Part 2: A Few Basic Problems
Intro: A Little Background
Part 1: Funny Things in High Dimensions


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Portrait Gallery



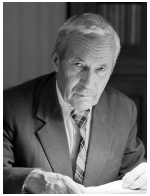
A. Boole Stott



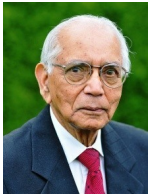
C. Stein



E. Wigner



A. Kolmogorov



C. R. Rao



R. Tibshirani



P. Buehlmann



S. van de Geer

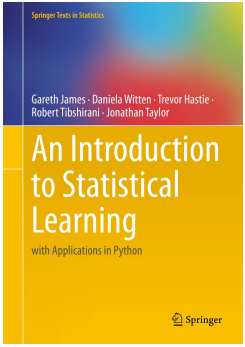


M. Wainwright

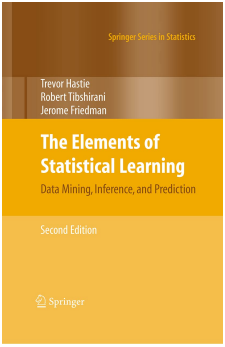


R. Vershynin

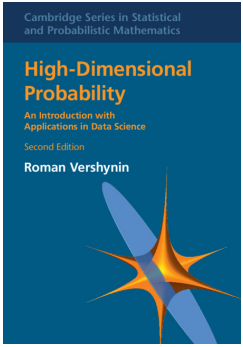
Books



James et al. (2023)



Hastie et al. (2009)



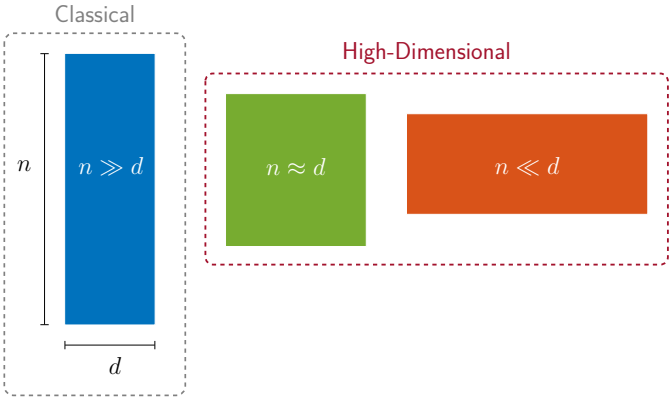
Vershynin (2026)

Key Questions

- Nowadays our datasets are *huge*: Does this imply we can always expect to *learn something* from them?
  - ... *What actually constitutes "a lot of data"?*
  - ... *Is having large datasets always a plus?*
- When can we expect to apply linear regression and get good results?
  - ... *And what does "good" mean?*
- Often, we impose many assumptions on our models and data: *Why*?
  - ... *Could we get rid of them?*
  - ... *And didn't deep learning solve this problem already?*

High-Dimensional Data

- We can roughly classify data in terms of "shape" when in tabular format:



A Word of Caution on Language

At times, we may use language that is a bit vague...

"x is big/small"

- **Contextual:** numbers are never "big" or "small" on their own, but *with respect to other numbers* that we take as reference
- **If it is unclear** what big/large/small refers to, **don't be afraid to ask**

"n or d (or both) go ($\to$) to ∞"

- "We treat the data **as if** n (or d, or both) are **essentially infinite**"
- This makes many simplifications possible (we will use "$\approx$" as a symbol)
- Without any added context, we should think of n and d as being *linked by some relationship* as they grow. Some examples:

$$d/n \to 0, \text{ or } d/n \to 0.2, \text{ or } d/\sqrt{n} \to 1, \text{ etc.}$$


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The HD Cube: Distance Grows

- Let's look at the distance between two points on the d-dimensional *hypercube*
  - $p_0 = (0, 0, \ldots, 0)^\top \in \mathbb{R}^d$ and $p_1 = (1, 1, \ldots, 1)^\top \in \mathbb{R}^d$

- $d = 2$: Square

$$\|d_1 - d_0\|_2 = \sqrt{(1-0)^2 - (1-0)^2} = \sqrt{2}$$

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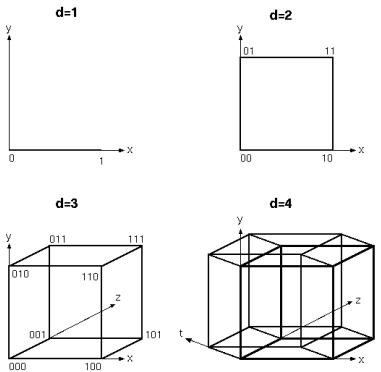
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$$\|d_1 - d_0\|_2 = \sqrt{\sum_{i=1}^{2} 1} = \sqrt{4}$$

...

- General d:

$$\|d_1 - d_0\|_2 = \sqrt{d} \quad \underset{d \to +\infty}{\longrightarrow} \quad +\infty$$



[Source: Wikimedia.org; Author: 4d44]

The HD Ball: Where is All the Volume?

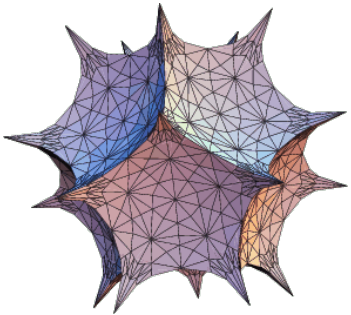
- What about the volume of the unit-radius, d-dimensional *hyper-ball*?
  - $\mathcal{B}^d := \left\{x \in \mathbb{R}^d \mid \|x\|_2 \leq 1\right\}$ and $\mathcal{V}_d = \text{volume}(\mathcal{B}^d)$

- $d = 2$: 2D Disk, $\mathcal{V}_2 = \pi \approx 3.142$
- $d = 3$: 3D Ball, $\mathcal{V}_3 = \frac{4}{3}\pi \approx 4.189$
- $d = 4$: 4D Ball, $\mathcal{V}_4 = \frac{\pi^2}{2} \approx 4.935$
- $d = 5$: $\mathcal{V}_5 = \frac{8\pi^2}{16} \approx 5.264$
- $d = 6$: $\mathcal{V}_6 = \frac{\pi^3}{6} \approx 5.168$
- $d = 7$: $\mathcal{V}_7 = \frac{16\pi^3}{105} \approx 4.725$

...

- General d:

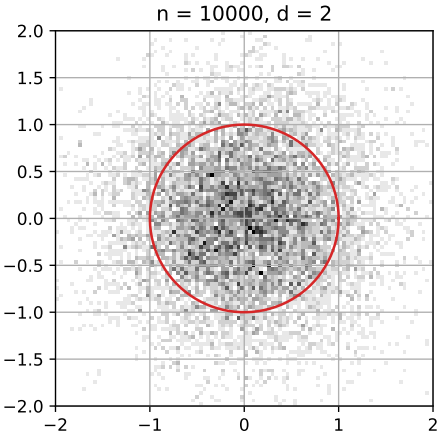
$$\mathcal{V}_d = \frac{\pi^{d/2}}{\Gamma(1 + d/2)} \quad \underset{d \to +\infty}{\longrightarrow} \quad 0$$



[Source: https://archive.lib.msu.edu/crcmath/math/math/h/h404.htm]

Example: Gaussian Concentration

Open in molab



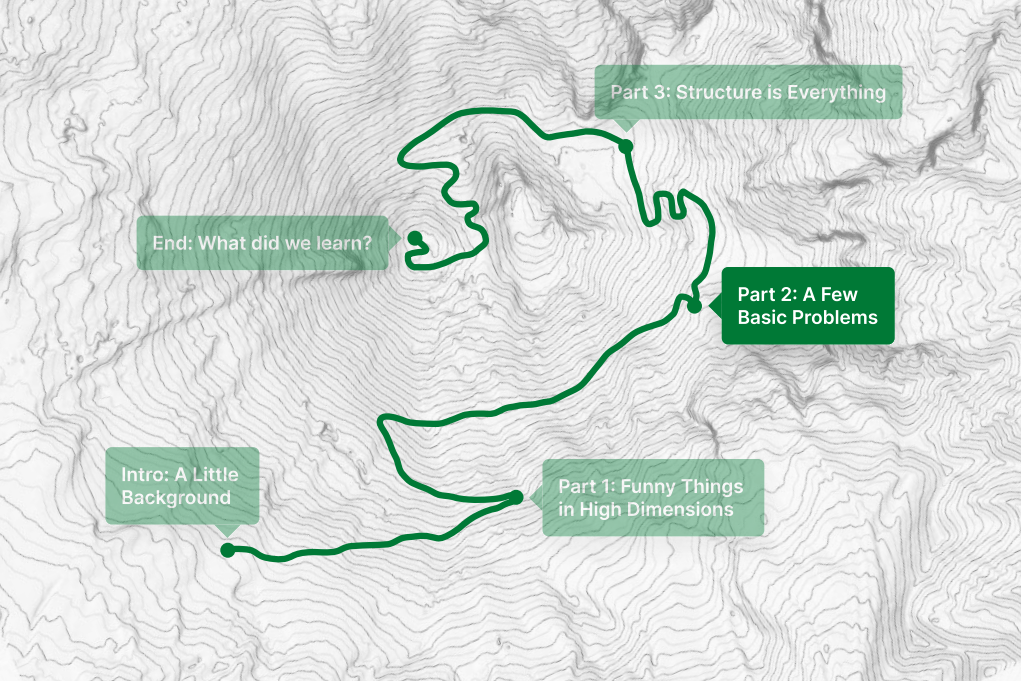
High-Dimensional Spaces in a Nutshell

- Summarizing what we have seen so far when $d \gg 3$:

1) There is **lots of room** in HD spaces: the diagonals of a hypercube grow as $\sqrt{d}$

2) If we work with constant distances (unit hyperball), then **volume is not strictly increasing**: in fact, it goes to 0 as d increases!

3) HD Gaussian data **concentrates around a spherical shell** as $d \to \infty$


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Estimating the Mean

Gaussian Location Problem

- Suppose we have $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2 \mathbb{I}_d)$: we want to estimate only μ
- An estimator is a function $\widehat{\mu} = \widehat{\mu}(X)$ of data $X := (X_1^\top, \dots, X_n^\top)^\top \in \mathbb{R}^{n \times d}$
- We consider the squared loss function

$$L(\widehat{\mu}, \mu) \;:=\; \|\widehat{\mu} - \mu\|_2^2 \,, \tag{Loss}$$

  where $\|\cdot\|_2$ is the Euclidean norm

- We want to construct $\widehat{\mu}$ to minimize the *risk*

$$\mathcal{R}(\widehat{\mu}, \mu) \;:=\; \mathbb{E}_X\big[L(\widehat{\mu}, \mu)\big] \;=\; \mathbb{E}_X\big[\|\widehat{\mu} - \mu\|_2^2\big] \tag{Risk}$$

  - Ideally, we pick $\widehat{\mu}$ such that, given *any other estimator* $\widetilde{\mu}$, it holds

$$\mathcal{R}(\widehat{\mu}, \mu) \;\leq\; \mathcal{R}(\widetilde{\mu}, \mu)$$

Quiz

- The *sample mean*, $\widehat{\mu}(X) = \frac{1}{n}\sum_{i=1}^{n} X_i$, is the natural choice

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Claim

Suppose that we have only 1 observation, X_1. The best estimator for μ in terms of squared loss risk is

$$\widehat{\mu}(X_1) \;=\; X_1,$$

regardless of the value of μ.

Quiz

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Claim

Suppose that we have only 1 observation, X_1. The best estimator for μ in terms of squared loss risk is

$$\widehat{\mu}(X_1) \;=\; X_1,$$

regardless of the value of μ.

Q: True or **false**?

James-Stein Estimator

- Willard James and Charles Stein showed in 1961 that ***it depends***

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- If $d \leq 2$, then indeed $\widehat{\mu}(X_1) = X_1$ does always achieve the minimum squared risk

- If $d > 2$, the James-Stein estimator

$$\widehat{\mu}^{\mathsf{JS}}(X_1) := \left(1 - \frac{(d-2)\sigma^2}{\|X_1\|_2^2}\right) X_1$$

has *strictly lower squared risk* than $\widehat{\mu}(X_1) = X_1$ for all $\mu \in \mathbb{R}^d$

  - This also generalizes to settings where $X_1 \sim \mathcal{N}(\mu, \Sigma)$ and to other loss functions

ESTIMATION WITH QUADRATIC LOSS

W. JAMES
FRESNO STATE COLLEGE
AND
CHARLES STEIN
STANFORD UNIVERSITY

1. Introduction

It has long been customary to measure the adequacy of an estimator by the smallness of its mean squared error. The least squares estimators were studied by Gauss and by other authors later in the nineteenth century. A proof that the best unbiased estimator of a linear function of the means of a set of observed random variables is the least squares estimator was given by Markov [12], a modified version of whose proof is given by David and Neyman [4]. A slightly more general theorem is given by Aitken [1]. Fisher [5] indicated that for large samples the maximum likelihood estimator approximately minimizes the mean squared error when compared with other reasonable estimators. This paper will be concerned with optimum properties or failure of optimum properties of the natural estimator in certain special problems with the risk usually measured by the mean squared error or, in the case of several parameters, by a quadratic function of the estimators. We shall first mention some recent papers on this subject and then give some results, mostly unpublished, in greater detail.

Pitman [13] in 1939 discussed the estimation of location and scale parameters and obtained the best estimator among those invariant under the affine transformations leaving the problem invariant. He considered various loss functions, in particular, mean squared error. Wald [18], also in 1939, in what may be considered the first paper on statistical decision theory, did the same for location parameters alone, and tried to show in his theorem 5 that the estimator obtained in this way is admissible, that is, that there is no estimator whose risk is no greater at any parameter point, and smaller at some point. However, his proof of theorem 5 is not convincing since he interchanges the order of integration in (30) without comment, and it is not clear that this integral is absolutely convergent. To our knowledge, no counterexample to this theorem is known, but in higher-dimensional cases, where the analogous argument seems at first glance only slightly less plausible, counterexamples are given in Blackwell [2] (which is discussed briefly at the end of section 3 of this paper) and in [14],

This work was supported in part by an ONR contract at Stanford University.

[James et al. (1961)]

Least Squares & Linear Regression

Setup

- Focus on a simple yet rich setting:

$$Y_i = \beta^\top X_i + \epsilon_i \tag{LM}$$

i.e. the **linear model**:

- $i \in \{1, \dots, n\}$, where integer $n > 0$ is the *sample size*
- $Y_i \in \mathbb{R}$ is the *response*
- $X_i \in \mathbb{R}^d$ are the *regressors/features*, where integer $d > 0$ is the *dimension*
- $\epsilon_i \in \mathbb{R}$ is the *noise/observation error*
- $\beta \in \mathbb{R}^d$ is the vector of *coefficients*

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  - $\epsilon_i \in \mathbb{R}$ is the *noise/observation error*
  - $\beta \in \mathbb{R}^d$ is the vector of *coefficients*

- We will assume that Y_i, X_i and ϵ_i are **random variables** (e.g. Gaussian) which are **identically and independently** (i.i.d.) generated according to (LM):
  - Although β is unknown, we assume (LM) is *true for the data*
  - Note that this is rarely the case... even though we can (almost) always run a linear regression on some dataset at hand!

Refresher

Notation for the linear regression model:

$$\boldsymbol{y} = \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{pmatrix}, \quad X = \begin{pmatrix} 1 & x_1^{(1)} & \dots & x_k^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(n)} & \dots & x_k^{(n)} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon^{(1)} \\ \vdots \\ \epsilon^{(n)} \end{pmatrix}$$

one has

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad \Rightarrow \quad \widehat{\beta} := (X^\top X)^{-1} X^\top Y$$

Today

We use capital letters, *drop the constant term*, and write instead

$$Y_i : \text{response}, \qquad X_i : \text{regressors/features}, \qquad \epsilon_i : \text{noise/error}$$

so that

$$Y_i = \beta^\top X_i + \epsilon_i \quad \Rightarrow \quad \widehat{\beta} = \Big(\tfrac{1}{n}\sum_{i=1}^n X_i X_i^\top\Big)^{-1}\Big(\tfrac{1}{n}\sum_{i=1}^n X_i Y_i\Big),$$

where (i) we divide both sides by n, (ii) use $X^\top X = \sum_{i=1}^n X_i X_i^\top$ (same for $X^\top Y$)

Linear Regression with $n > d$

- Let now n be a large number:

$$Y_i = \beta^\top X_i + \epsilon_i \quad \rightarrow \quad (Y_1, X_1),\ (Y_2, X_2),\ \ldots,\ (Y_n, X_n)$$

- We now are in a more useful case where $\beta \in \mathbb{R}^d$, $d > 1$ and $n > d$
- How to estimate β?

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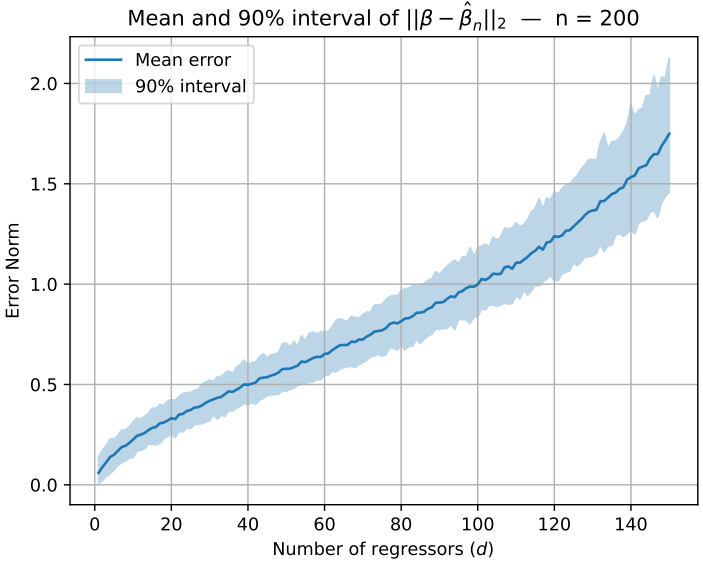
- We now are in a more useful case where $\beta \in \mathbb{R}^d$, $d > 1$ and $n > d$
- How to estimate β?
  - Linear regression:

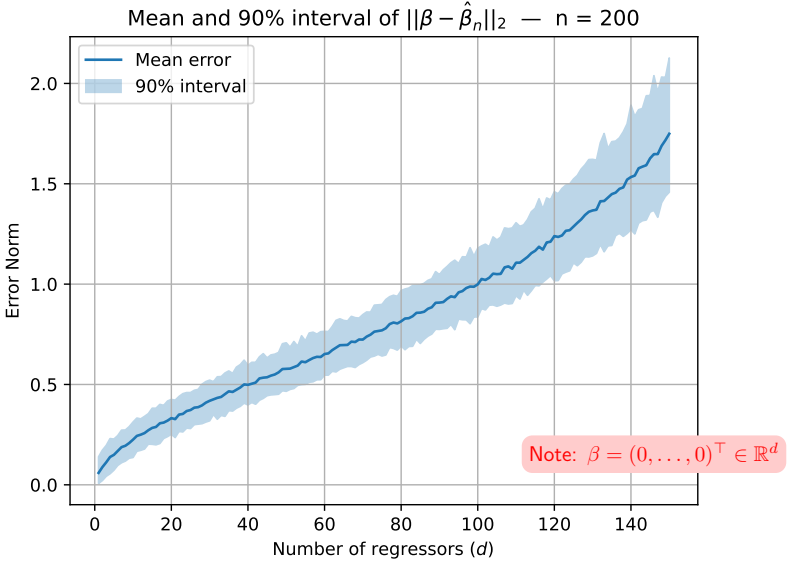
$$\widehat{\beta}_n = \left(\tfrac{1}{n}\sum_{i=1}^n X_i X_i^\top\right)^{-1}\left(\tfrac{1}{n}\sum_{i=1}^n X_i Y_i\right) = \beta + \widehat{\Sigma}_n^{-1}\left(\tfrac{1}{n}\sum_{i=1}^n X_i \epsilon_i\right)$$

    where we set $\widehat{\Sigma}_n := \frac{1}{n}\sum_{i=1}^n X_i X_i^\top$
  - Since d is small relative to n,

$$\tfrac{1}{n}\sum_{i=1}^n X_i \epsilon_i \approx \mathbb{E}[X_i \epsilon_i] \quad \text{and} \quad \widehat{\Sigma}_n \approx \mathbb{E}[X_i X_i^\top]$$

  - Assuming $\mathbb{E}[X_i \epsilon_i] = 0$ and $\mathbb{E}[X_i X_i^\top]$ invertible, we have $\widehat{\beta}_n \approx \beta$


Mean and 90% interval of $||\beta - \hat{\beta}_n||_2$ — n = 200
Mean error
90% interval
Error Norm
0.0
0.5
1.0
1.5
2.0
0
20
40
60
80
100
120
140
Number of regressors (d)


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Number of regressors (d)
Note: $\beta = (0, \ldots, 0)^\top \in \mathbb{R}^d$

Linear regression with $n \approx d$

- Everything like the previous slide, but what if d is almost as large as n, say

$$d = 0.8\, n$$

- We can still run a linear regression, since $n > d$

Linear regression with $n \approx d$

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$$d = 0.8\,n$$

- We can still run a linear regression, since $n > d$
- But let's think about the matrix we have to invert:

$$\widehat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top \in \mathbb{R}^{d\times d} \quad \xrightarrow{\text{estimates}} \quad \mathbb{E}[X_i X_i^\top] \quad \xrightarrow{\text{symmetric}} \quad \frac{(d-1)d}{2} \text{ coeff's}$$

  - When $n > 3$, so that $(0.8n - 1) > n/2$, there are more than

$$n^2/4$$

    coefficients to estimate with **an average over only n data points**!
  - To have a "fair" regression problem, it seems we can have *at most* $d \approx \sqrt{n}$

Covariance Estimation Problem

- Focus on comparing

$$\widehat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^n X_i X_i^\top \in \mathbb{R}^{d\times d} \quad \text{vs.} \quad \Sigma = \mathbb{E}[X_i X_i^\top]$$

- We want to know what happens when the **aspect ratio**

$$\alpha = \frac{d}{n}$$

is not small, i.e. $\alpha \not\approx 0$.

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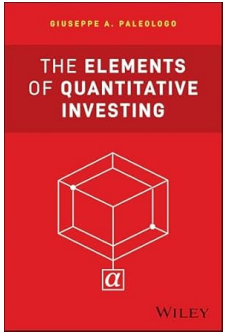
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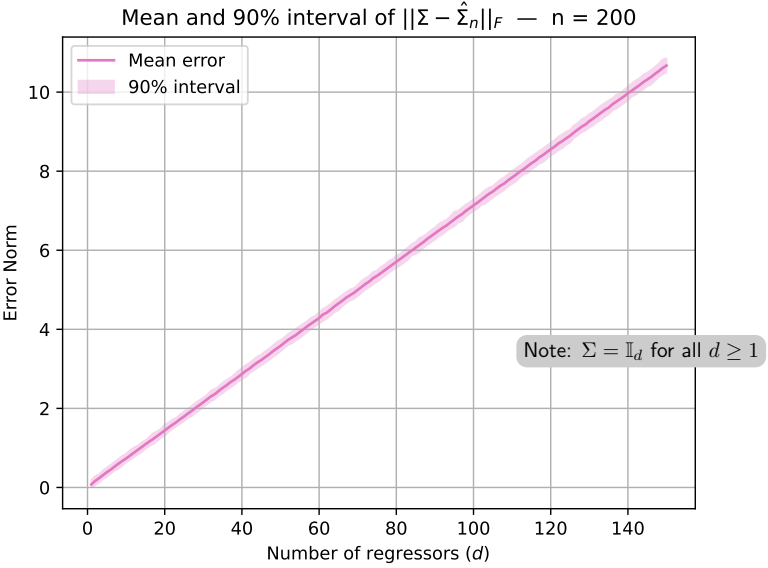
$$\alpha = \frac{d}{n}$$

  is not small, i.e. $\alpha \not\approx 0$.

- In finance, covariance estimation is hugely relevant!
- Comes into play when doing *minimum-variance portfolio* analysis; a "pre-step" for most models
- Simple simulation:
  - $X_i \sim \mathcal{N}(0, \mathbb{I}_d)$ so that $\Sigma = \mathbb{I}_d$
  - Is $\widehat{\Sigma}_n$ any similar to Σ if e.g. $\alpha = 0.3$?

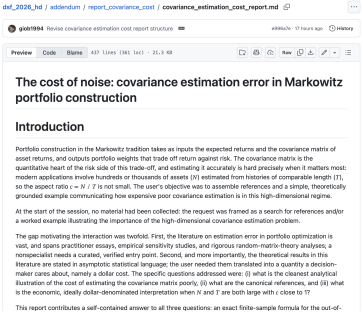


[See also: Paleologo (2025); https://osquant.com/papers/a-quants-guide-to-covariance-matrix-estimation/]


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90% interval
Error Norm
0
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10
0
20
40
60
80
100
120
140
Number of regressors (d)
Note: $\Sigma = \mathbb{I}_d$ for all $d \geq 1$

Financial Costs of HD Covariance Estimation

- Back-of-the-envelope cost analysis for classical Markowitz portfolio theory
  - Very informal; 2 prompts with GLM 5.3
- Simplified setting:
  - Gaussian i.i.d. returns X_i, $\Sigma = \mathbb{I}_d$
  - Equal expected returns across assets
  - Compare using Σ vs. $\widehat{\Sigma}_n$
- Assume \$1 billion invested
  - Investor is risk averse, etc.
  - Equal portfolio weights over all assets
- Depending on α...
  - $\alpha = 0.5 \;\Rightarrow\;$ \$6.4 million ann. cost
  - $\alpha = 0.95 \;\Rightarrow\;$ \$121.6 million ann. cost


dsf_2026_hd / addendum / report_covariance_cost / covariance_estimation_cost_report.md

giobi1994 Revise covariance estimation cost report structure — e9984a7e · 17 hours ago · History

Preview | Code | Blame | 437 Lines (361 loc) · 21.3 KB | Raw

The cost of noise: covariance estimation error in Markowitz portfolio construction

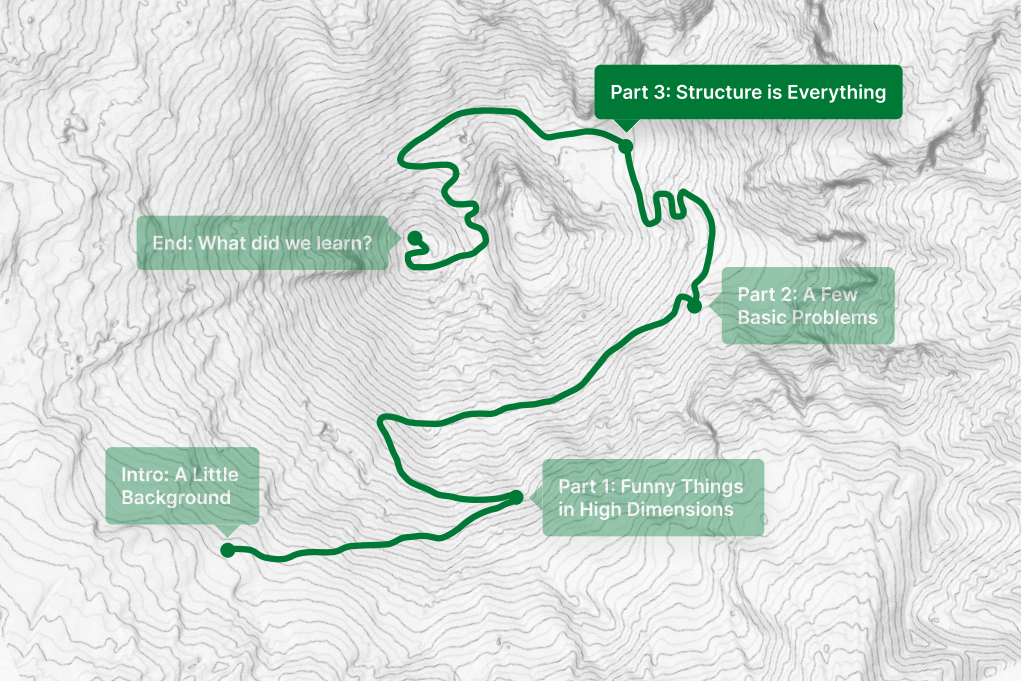
Introduction

Portfolio construction in the Markowitz tradition takes as inputs the expected returns and the covariance matrix of asset returns, and outputs portfolio weights that trade off return against risk. The covariance matrix is the quantitative heart of the risk side of this trade-off, and estimating it accurately is hard precisely when it matters most: modern applications involve hundreds or thousands of assets (N) estimated from histories of comparable length (T), so the aspect ratio $c = N / T$ is not small. The user's objective was to assemble references and a simple, theoretically grounded example communicating how expensive poor covariance estimation is in this high-dimensional regime.

At the start of the session, no material had been collected: the request was framed as a search for references and/or a worked example illustrating the importance of the high-dimensional covariance estimation problem.

The gap motivating the interaction was twofold. First, the literature on estimation error in portfolio optimization is vast, and spans practitioner essays, empirical sensitivity studies, and rigorous random-matrix-theory analyses; a nonspecialist needs a curated, verified entry point. Second, and more importantly, the theoretical results in this literature are stated in asymptotic statistical language; the user needed them translated into a quantity a decision-maker cares about, namely a dollar cost. The specific questions addressed were: (i) what is the cleanest analytical illustration of the cost of estimating the covariance matrix poorly, (ii) what are the canonical references, and (iii) what is the economic, ideally dollar-denominated interpretation when N and T are both large with c close to 1?

This report contributes a self-contained answer to all three questions: an exact finite-sample formula for the out-of-


Part 3: Structure is Everything
End: What did we learn?
Part 2: A Few Basic Problems
Intro: A Little Background
Part 1: Funny Things in High Dimensions

Structure is Everything

- Why did our naive linear regression approach fail in high dimensions?
- Because HD spaces are **very big**!
  - Estimating β with least squares effectively means searching over $\mathbb{R}^d$ with little-to-no information when $\alpha \not\approx 0$
  - Same argument for Γ when doing covariance estimation

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- An informal ansatz would be:

  "If α is non-negligible (say, > 0.2), don't trust linear regression"

- Is this really all we can do?

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- An informal ansatz would be:

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- Is this really all we can do?
  - Not enough to solely rely on information from the data...
  - But we can *add some information* of our own!

    "**Add reasonable structure** to the problem"
  - We call this private info a **prior**

[See also: Efron (1986)]

Possible Priors

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(A) "Most coefficients in β are *close to* 0"

$$\|\beta\|_2 := \sqrt{\sum_{j=1}^{d} \beta_j^2} \text{ small } \rightarrow \text{ penalize large } \|\beta\|_2 \rightarrow \text{ Ridge regression}$$

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(B) "Most coefficients in β are *exactly* zero"

$$\|\beta\|_0 := \sum_{j=1}^d 1\{\beta_j \neq 0\} \text{ small } \rightarrow \text{ penalize large } \|\beta\|_0 : \textit{ too hard to compute!}$$
$$\rightarrow \text{ use } \|\beta\|_1 := \sum_{j=1}^d |\beta_j| \text{ instead}$$
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$$\rightarrow \text{ LASSO}$$

(C) "Features in X_i are partly redundant"

Reduce X_i along principal components (PCA) $\rightarrow$ PC regression

Ridge Regression

Ridge Regression I

- **Ridge regression (RR)** has a long history (see Hastie, 2020)
  - 1940s (Andrey Tikhonov) and then later 1970s (Arthur E. Hoerl)
- Find $\widehat{\beta}_n^{\text{RR}}(\lambda)$, where $\lambda \in [0, \infty)$, to solve (Hastie et al., 2009, Ch.3.4.1)

$$\min_{\beta \in \mathbb{R}^d} \underbrace{\frac{1}{n} \sum_{i=1}^{n} (Y_i - \beta^\top X_i)^2}_{\text{Least squares loss}} + \underbrace{\lambda \sum_{i=1}^{d} \beta_j^2}_{\text{Ridge penalty}}$$

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- We can write the solution (after some linear algebra) as

$$\widehat{\beta}_n^{\text{RR}}(\lambda) = \left(\tfrac{1}{n} \textstyle\sum_{i=1}^{n} X_i X_i^\top + \lambda \mathbb{I}_d\right)^{-1} \left(\tfrac{1}{n} \textstyle\sum_{i=1}^{n} X_i Y_i\right)$$

  - The explicit formula is not hard to compute: RR is a quadratic problem, like OLS
  - Just need to note that $\sum_{i=1}^{d} \beta_j^2 = \beta^\top \beta$
  - Can use, e.g., `linear_model.Ridge` from `scikit-learn`

Ridge Regression II

- Note that

$$\widehat{\beta}_n^{\mathsf{RR}}(\lambda) = \big(\tfrac{1}{n}\sum_{i=1}^n X_i X_i^\top + \lambda \mathbb{I}_d\big)^{-1}\big(\tfrac{1}{n}\sum_{i=1}^n X_i Y_i\big)$$
$$= \beta - \lambda\big(\widehat{\Sigma}_n + \lambda \mathbb{I}_d\big)^{-1}\beta + \big(\widehat{\Sigma}_n + \lambda \mathbb{I}_d\big)^{-1}\big(\tfrac{1}{n}\sum_{i=1}^n X_i \epsilon_i\big)$$

  - Recall: $\widehat{\Sigma}_n := \frac{1}{n}\sum_{i=1}^n X_i X_i^\top$
  - We also used that $\frac{1}{n}\sum_{i=1}^n X_i Y_i = \frac{1}{n}\sum_{i=1}^n (X_i X_i^\top \beta + \epsilon_i)$ can be written as
$$\tfrac{1}{n}\sum_{i=1}^n (\widehat{\Sigma}_n \beta + \lambda \mathbb{I}_d)\beta - \tfrac{1}{n}\sum_{i=1}^n (\lambda \mathbb{I}_d)\beta + \tfrac{1}{n}\sum_{i=1}^n (\widehat{\Sigma}_n \beta + \epsilon_i)$$

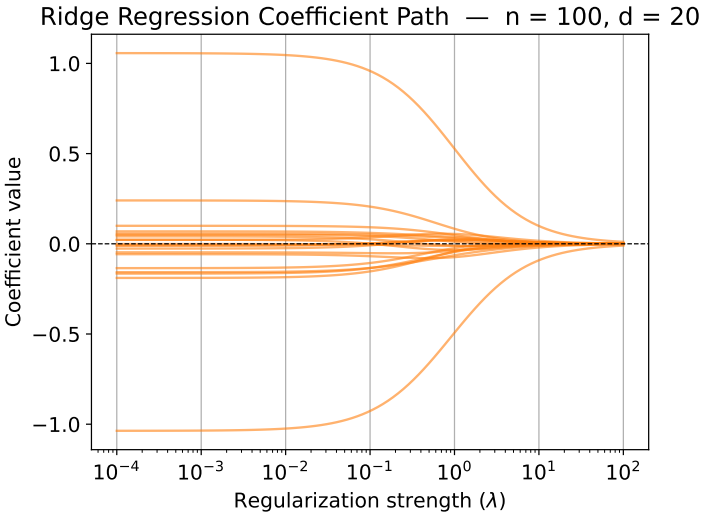
Ridge Regression II

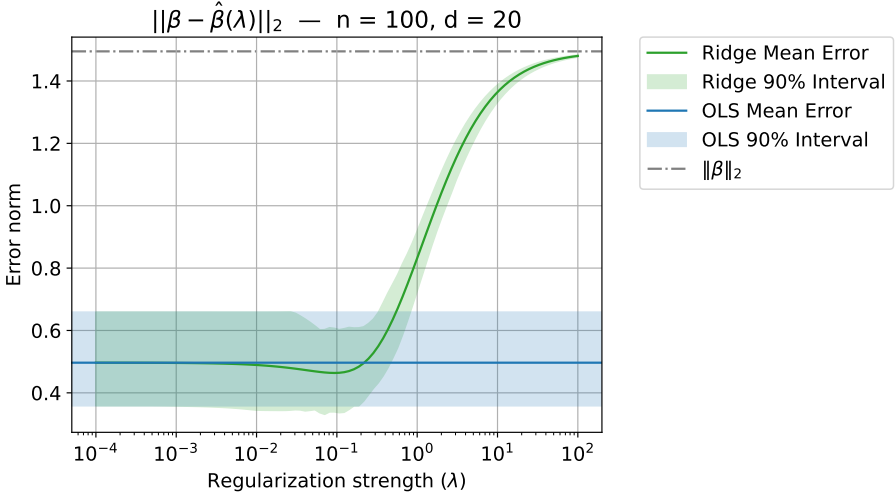
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- If $\lambda \to \infty$, then $\|\widehat{\beta}_n^{\mathsf{RR}}(\lambda)\|_2 \to 0$
  - Large choices of λ reduce the variability of estimator $\widehat{\beta}_n^{\mathsf{RR}}(\lambda)$...
  - But they also push it towards 0!
- Ridge regression should work well when "β is close to 0"
  - In practice, RR is widely used to help with situations where the aspect ratio α is less-than-ideal, and in neural networks to regularize learning


Ridge Regression Coefficient Path — n = 100, d = 20
Coefficient value
1.0
0.5
0.0
−0.5
−1.0
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)


$||\beta - \hat{\beta}(\lambda)||_2$ — n = 100, d = 20
Error norm
1.4
1.2
1.0
0.8
0.6
0.4
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)
Ridge Mean Error
Ridge 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$

Ridge Regression in HD

- What happens when we use RR in the HD setting?
- A thing to note is that RR "always works" if $\lambda > 0$:

$$\left(\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{\widehat{\Sigma}_n} + \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} (1+\lambda)^{-1} & 0 \\ 0 & \lambda^{-1} \end{bmatrix}$$

  - In this case, $\widehat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top$ is not even invertible; no proper unique LS solution!

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  - The *minimum-norm least squares* solution can, however, always be defined:

$$\widehat{\beta}^{\mathsf{MLS}} := \arg\min_{\beta \in \mathbb{R}^d} n^{-1} \sum_{i=1}^n (Y_i - \beta^\top X_i)^2$$

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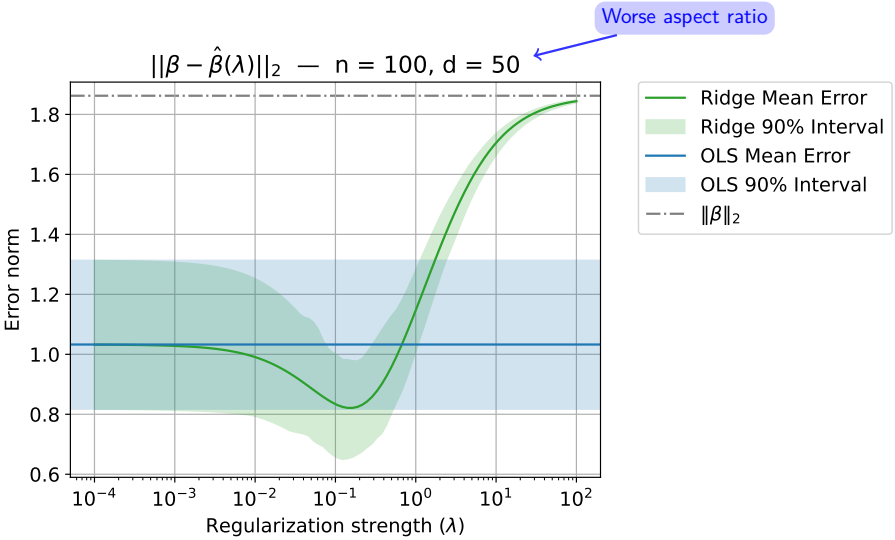
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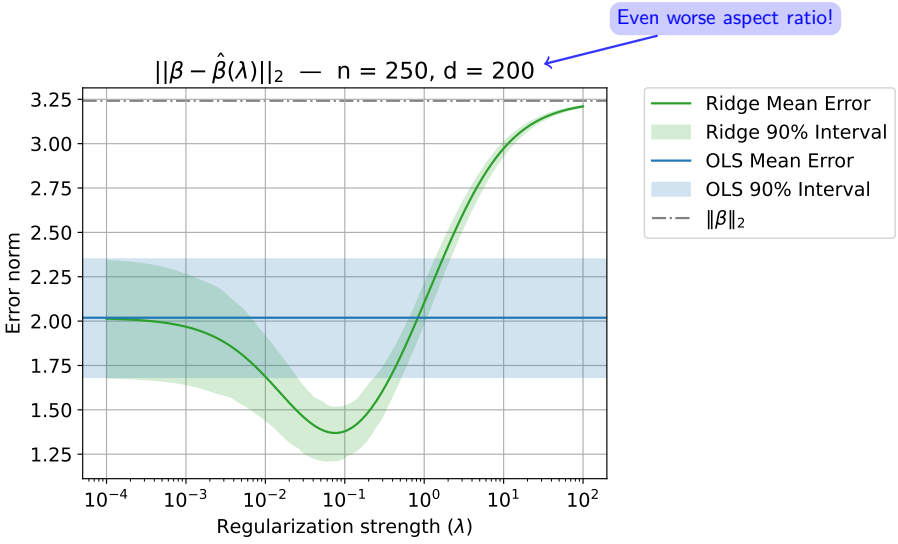
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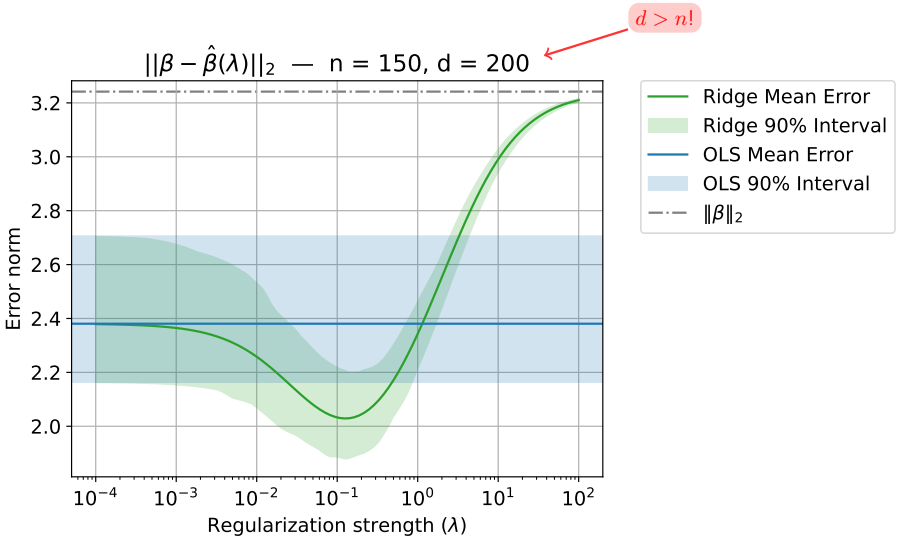
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- One can use RR when $d \ll n$, $d \approx n$, or even $d \gg n$
  - In fact, when we have a non-trivial aspect ratio ($d/n \to \alpha > 0$ as d, n grow), $\widehat{\beta}_n^{\mathsf{RR}}$ also works well in a "factor model" setting (will see in a few slides)
  - Does not mean $\widehat{\beta}_n^{\mathsf{RR}}$ is necessarily a *good* estimate of the true β

[See also: De Mol et al. (2024)]


Worse aspect ratio
$||\beta - \hat{\beta}(\lambda)||_2$ — n = 100, d = 50
Ridge Mean Error
Ridge 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$
Error norm
1.8
1.6
1.4
1.2
1.0
0.8
0.6
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)


Even worse aspect ratio!
$||\beta - \hat{\beta}(\lambda)||_2$ — n = 250, d = 200
Error norm
Regularization strength (λ)
Ridge Mean Error
Ridge 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$


$d > n!$
$||\beta - \hat{\beta}(\lambda)||_2$ — n = 150, d = 200
Error norm
Regularization strength (λ)
Ridge Mean Error
Ridge 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$

LASSO

LASSO I

- **LASSO** takes a more structured approach and assumes **sparsity**, i.e. most coefficients in β are simply 0, so we could just "clip them off"
  - Popularized by Robert Tibshirani in 1996, but goes back to the '80s
- We can count the number of non-zero components with

$$\|\beta\|_0 := \sum_{j=1}^{d} 1\{\beta_j \neq 0\}$$

  - Unfortunately, $\|\cdot\|_0$ is too hard to compute and optimize with (need to count all possible subsets of coefficients...)

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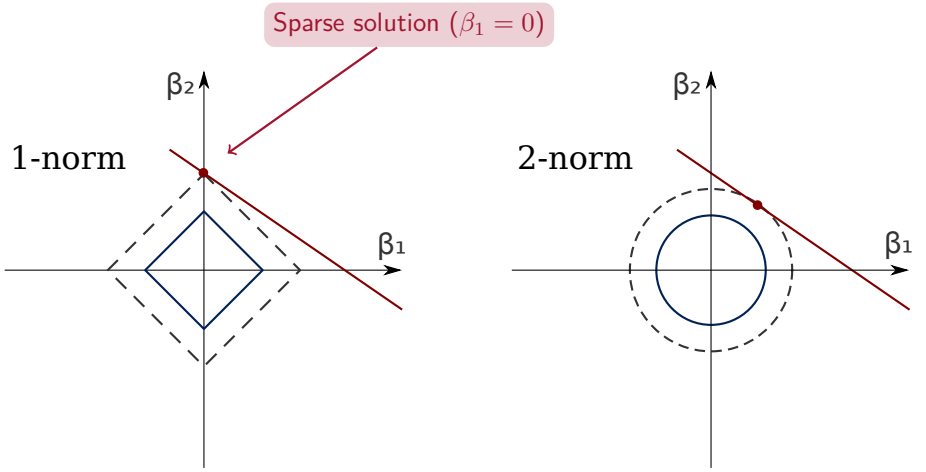
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  - Unfortunately, $\|\cdot\|_0$ is too hard to compute and optimize with (need to count all possible subsets of coefficients...)
- Use instead the 1-norm and get $\widehat{\beta}_n^{\mathsf{L}}(\lambda)$ by solving (Hastie et al., 2009, Ch.3.4.2)

$$\min_{\beta \in \mathbb{R}^d} \underbrace{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \beta^\top X_i)^2}_{\text{Least squares loss}} + \underbrace{\lambda \sum_{i=1}^{d}|\beta_j|}_{\text{LASSO penalty}}$$

LASSO II



[Source: Wikimedia.org; Author: Nicoguaro]

LASSO III

- Unfortunately, there is no explicit formula for $\widehat{\beta}_n^{\mathsf{L}}(\lambda)$!
  - However, the LASSO estimator is easily solvable *numerically*
  - E.g., `linear_model.Lasso` from `scikit-learn`
  - Very fast solvers available [for Python, see `celer` by Massias et al. (2018)]

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  - Very fast solvers available [for Python, see `celer` by Massias et al. (2018)]
- LASSO has both nice and less nice properties

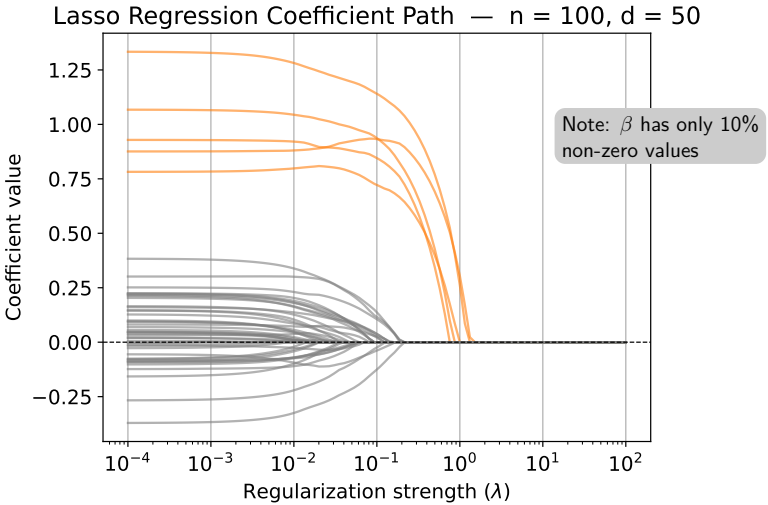
  (+) Consistent subset selection (under appropriate technical assumptions),

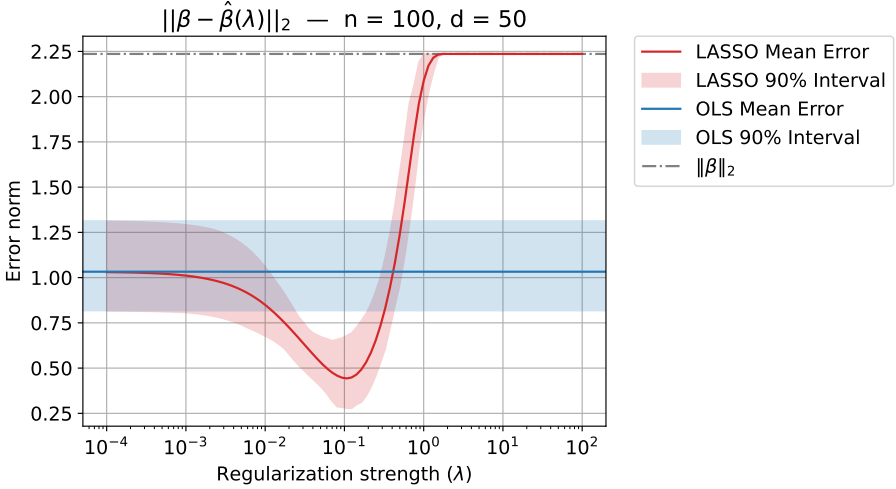
$$\mathbb{P}\big(\#\widehat{\beta}^{\mathsf{L}}_n(\lambda) \subseteq \#\beta\big) \underset{n,d\to+\infty}{\longrightarrow} 1,$$

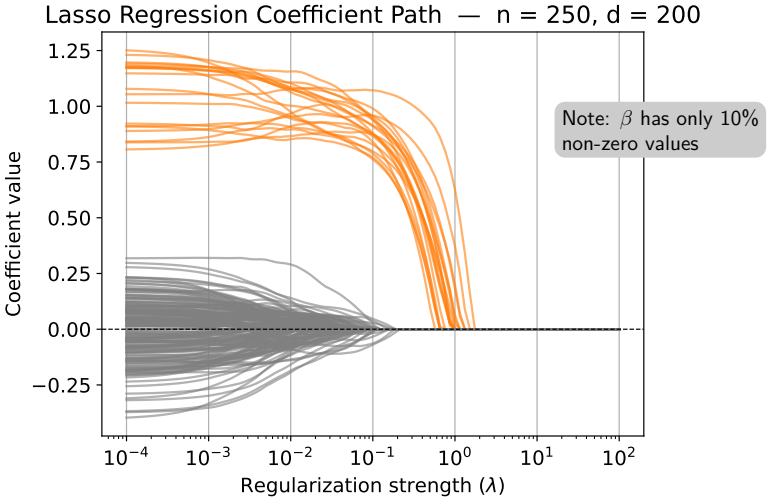
  where $\#\beta := \{\, j \,|\, 1 \leq j \leq d, \beta_j \neq 0\}$ and similary for $\#\widehat{\beta}^{\mathsf{L}}_n(\lambda)$

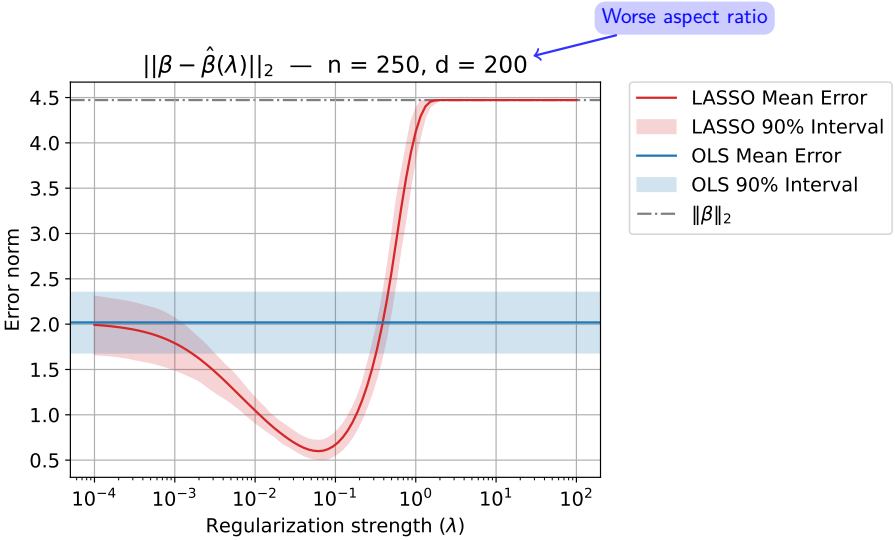
  (–) Under more "realistic" conditions (e.g., strong linear dependence of regressors), vanilla lasso often *does not* correctly select regressors with non-zero coefficients
- Many LASSO variants/extensions exist: Adaptive LASSO, Group LASSO, SCAD, Fused LASSO, Square-root LASSO, Debiased LASSO...

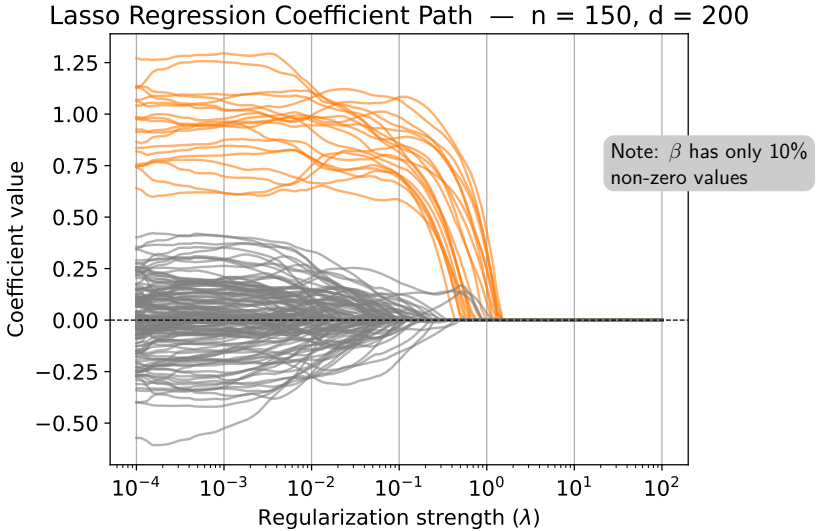
[See also: Freijeiro-González et al. (2022)]

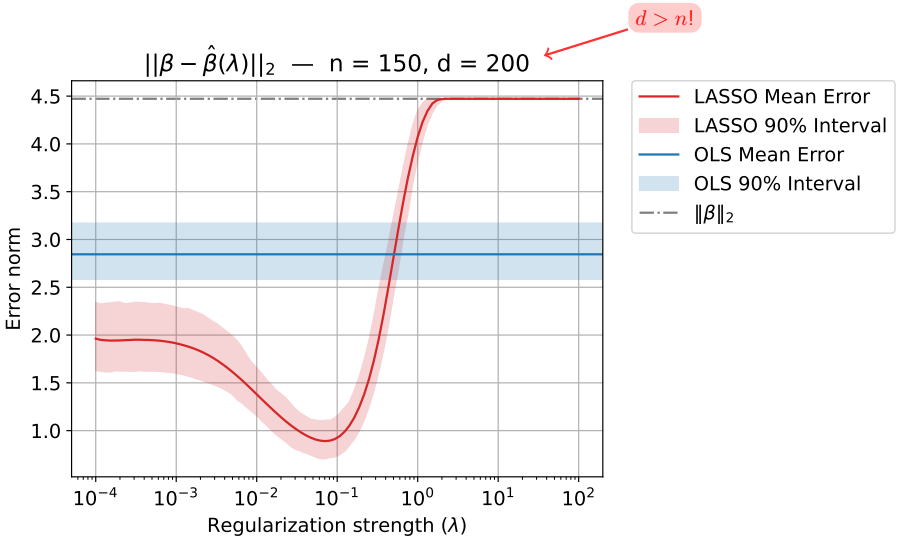

Lasso Regression Coefficient Path — n = 100, d = 50
Coefficient value
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)
Note: β has only 10% non-zero values


$||\beta - \hat{\beta}(\lambda)||_2$ — n = 100, d = 50
Error norm
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)
LASSO Mean Error
LASSO 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$

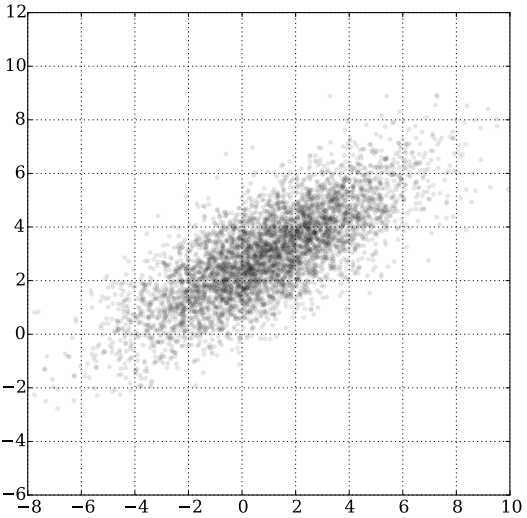

Lasso Regression Coefficient Path — n = 250, d = 200
Coefficient value
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)
Note: β has only 10% non-zero values

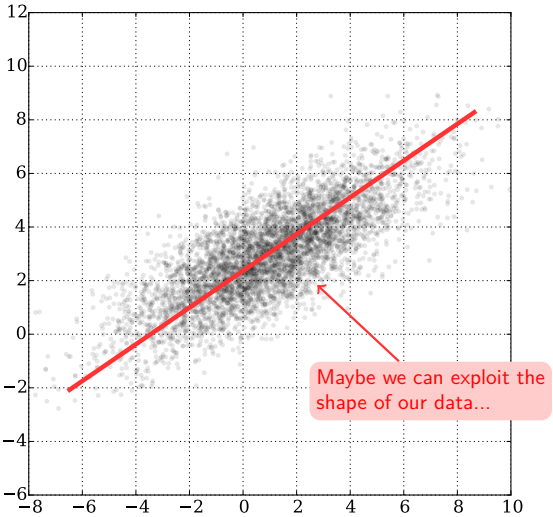

Worse aspect ratio
$||\beta - \hat{\beta}(\lambda)||_2$ — n = 250, d = 200
LASSO Mean Error
LASSO 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$
Error norm
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)


Lasso Regression Coefficient Path — n = 150, d = 200
Coefficient value
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Regularization strength (λ)
Note: β has only 10% non-zero values


$d > n!$
$||\beta - \hat{\beta}(\lambda)||_2$ — n = 150, d = 200
Error norm
Regularization strength (λ)
LASSO Mean Error
LASSO 90% Interval
OLS Mean Error
OLS 90% Interval
$\|\beta\|_2$

Principal Component Regression


12
10
8
6
4
2
0
−2
−4
−6
−8
−6
−4
−2
0
2
4
6
8
10


12
10
8
6
4
2
0
−2
−4
−6
−8
−6
−4
−2
0
2
4
6
8
10
Maybe we can exploit the
shape of our data...

Principal Component Analysis

- We graphically identified the direction onto which data is **"most spread"**
- Can we do this in a data-driven way?

Principal Component Analysis

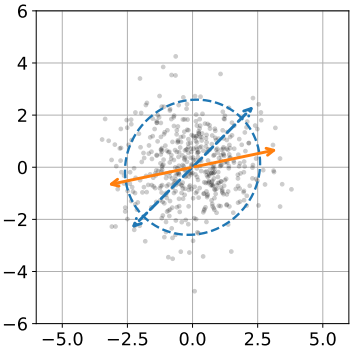
- We graphically identified the direction onto which data is **"most spread"**
- Can we do this in a data-driven way?
- The (linear) way to do it is by using **Principal Component Analysis (PCA)**

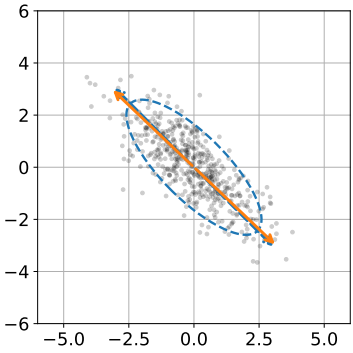
$$\widehat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top \quad \overset{\text{SVD}}{\longrightarrow} \quad U_n S_n U_n^\top \quad \overset{\text{principal comp.}}{\longrightarrow} \quad u_1,\ [S_n]_{11} =: s_1$$

  - Singular Value Decomposition (SVD) is a way to extract the directions of "principal variation" of data covariance matrix $\widehat{\Sigma}_n$ (assuming X_i's are centered)
  - u_1 is the (first) **principal component** (*direction* of component, $\|u_1\|_2 = 1$)
  - s_1 is the (first) **singular value** (*size* of component, $s_1 \geq 0$)
  - Can generalize this to all $j = 1, 2, \ldots, d$ and get component pairs (u_j, s_j)
  - See `decomposition.PCA` from `scikit-learn`

Example: Gaussian Data PCA

Open in molab





Principal Component Regression I

- Suppose we want to keep only one data direction, u_1
- Linear algebra of projections tells us we can do it easily:

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- In general, assume we keep only the first k data PCs:
  (1) Use SVD to get U_n for $\widehat{\Gamma}_n$
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- **Principal Component Regression (PCR)** is (Hastie et al., 2009, Ch.3.5.1)

$$\begin{aligned}\widehat{\beta}_{n,k}^{\text{PC}} &= \left(\tfrac{1}{n}\textstyle\sum_{i=1}^n Z_i Z_i^\top\right)^{-1}\left(\tfrac{1}{n}\textstyle\sum_{i=1}^n Z_i Y_i\right) \\ &= \left(\tfrac{1}{n}\textstyle\sum_{i=1}^n P_k^\top X_i X_i^\top P_k\right)^{-1}\left(\tfrac{1}{n}\textstyle\sum_{i=1}^n P_k^\top X_i Y_i\right)\end{aligned}$$

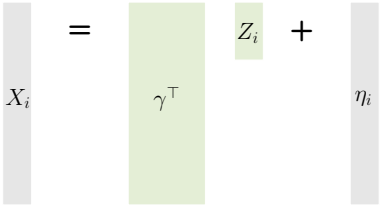
Principal Component Regression II

- With PCR, we certainly "throw away" some information by choosing $k < d$: **When does this pay off?**
- Suppose

$$Y_i = \beta^\top F_i + \epsilon_i$$
$$X_i = \gamma^\top F_i + \eta_i, \qquad \text{(Factor Model)}$$

where $\gamma \in \mathbb{R}^{k \times d}$, $F_i \in \mathbb{R}^k$ and $\eta_i \in \mathbb{R}^d$ is noise independent of ϵ_i

  - The elements of F_i are commonly called **factors**



Principal Component Regression III

- Suppose we know k for simplicity, that $F_i \in \mathbb{R}^k$ and $\eta_i \in \mathbb{R}^d$ have no correlation, and also that their components are never collinear
- With a little algebra, one can show

$$\Sigma = \mathbb{E}[X_i X_i^\top] = \gamma^\top \mathbb{E}[F_i F_i^\top]\gamma + \mathbb{E}[\eta_i \eta_i^\top]$$

and, importantly,

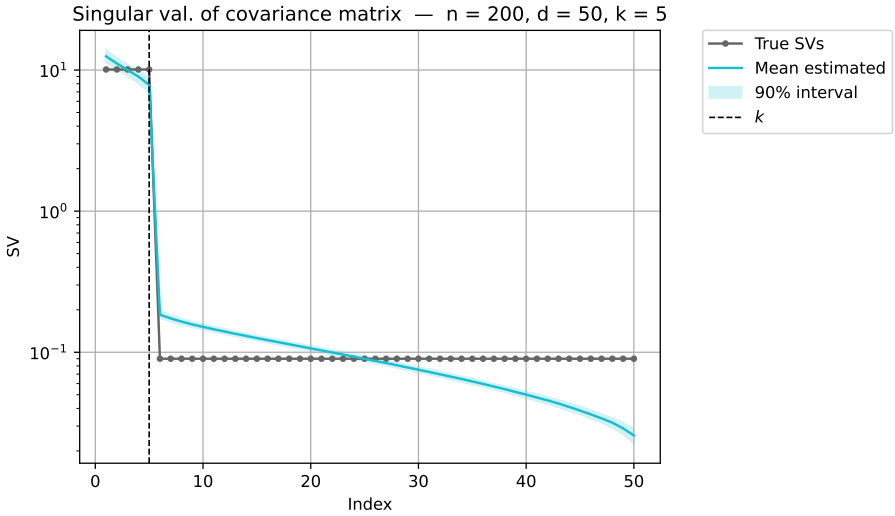
- $\gamma^\top \mathbb{E}[F_i F_i^\top]\gamma \in \mathbb{R}^d$ has only k non-zero singular values
- $\mathbb{E}[\eta_i \eta_i^\top] \in \mathbb{R}^d$ has d non-zero singular values
- If k is fixed, but $d \to \infty$, then Σ will have exactly k singular values $(s_1, \ldots, s_k)$ also diverging to infinity

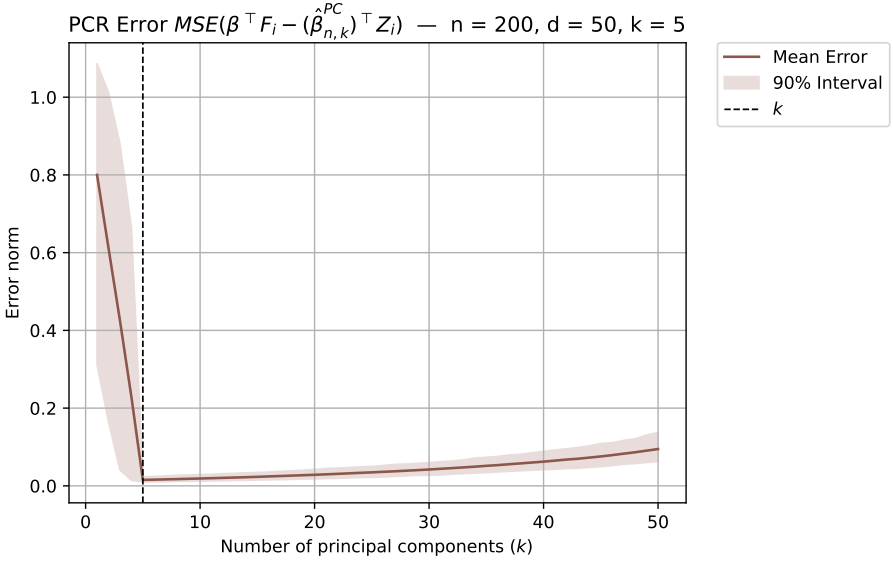
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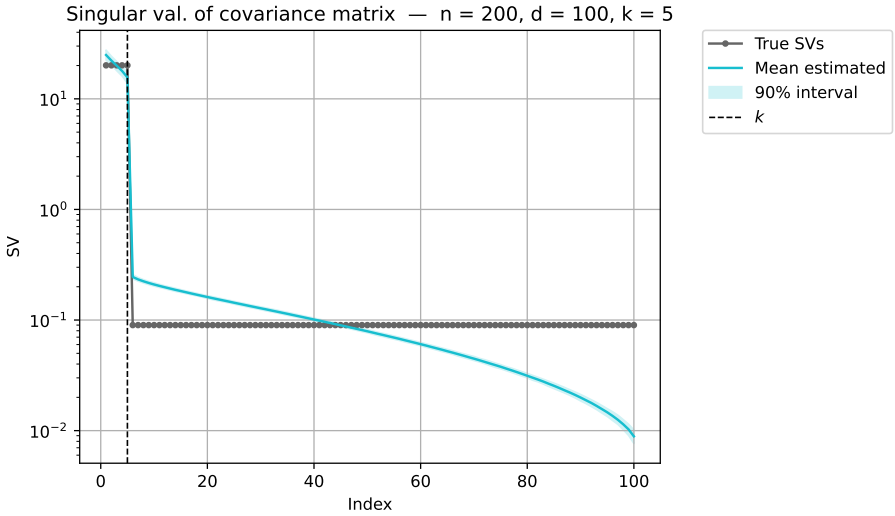
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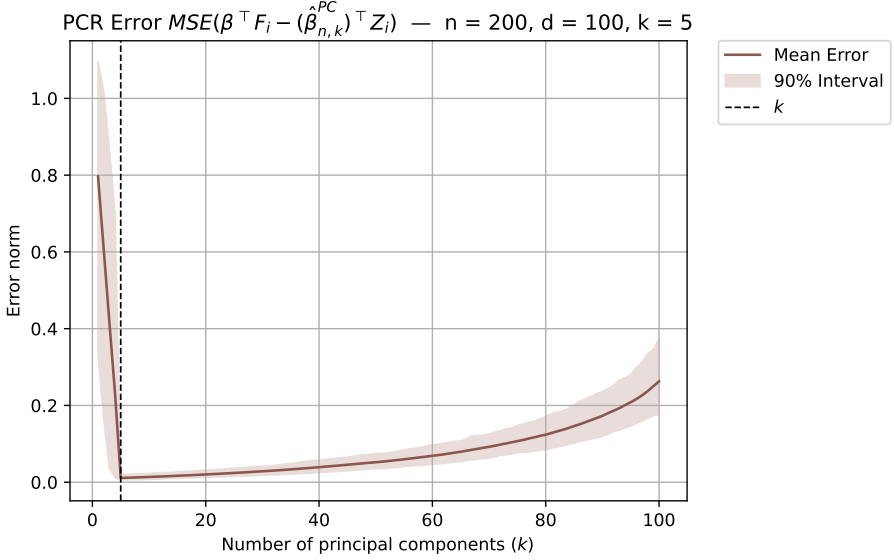
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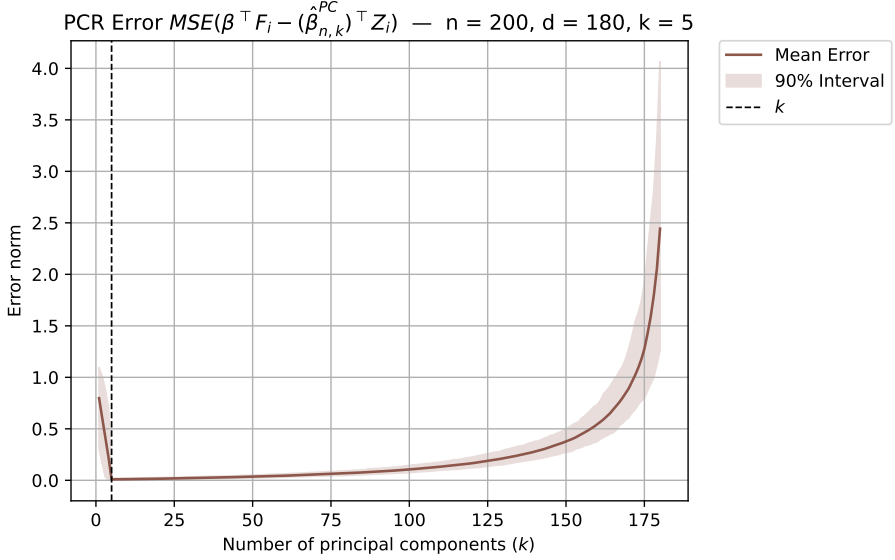
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  - If k is fixed, but $d \to \infty$, then Σ will have exactly k singular values $(s_1, \ldots, s_k)$ also diverging to infinity
- If the data comes from a factor model, PCA actually *skims out* the "diverging" components (since the data gets spread over these), and hence we **can recover the underlying (unobserved) factors**
  - The remainder PCs all come from noise η_i, so we can just throw them away!

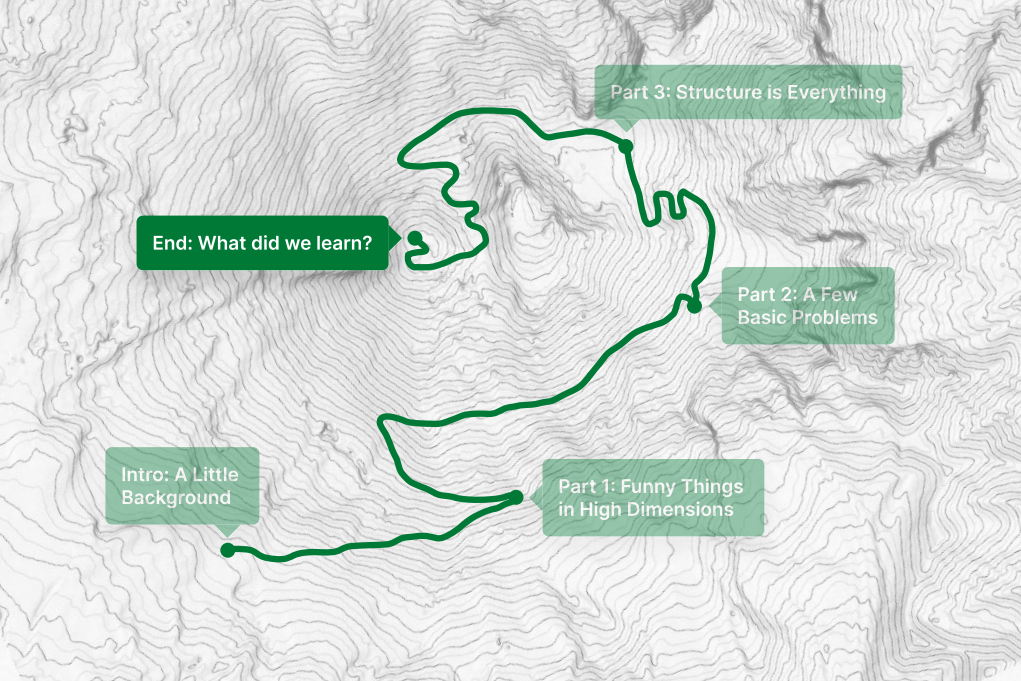

Singular val. of covariance matrix — n = 200, d = 50, k = 5
True SVs
Mean estimated
90% interval
k
SV
Index
10^1
10^0
10^{-1}
0
10
20
30
40
50


PCR Error $MSE(\beta^\top F_i - (\hat{\beta}^{PC}_{n,k})^\top Z_i)$ — n = 200, d = 50, k = 5
Mean Error
90% Interval
k
Error norm
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
Number of principal components (k)


Singular val. of covariance matrix — n = 200, d = 100, k = 5
True SVs
Mean estimated
90% interval
k
SV
Index
10^{1}
10^{0}
10^{-1}
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0
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40
60
80
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20
40
60
80
100
Number of principal components (k)


PCR Error $MSE(\beta^\top F_i - (\hat{\beta}^{PC}_{n,k})^\top Z_i)$ — n = 200, d = 180, k = 5
Mean Error
90% Interval
k
Error norm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
25
50
75
100
125
150
175
Number of principal components (k)


Part 3: Structure is Everything
End: What did we learn?
Part 2: A Few Basic Problems
Intro: A Little Background
Part 1: Funny Things in High Dimensions

Key Questions, Reprise

- Nowadays our datasets are *huge*: Does this imply we can always expect to *learn something* from them?
  - ✗ **Not always**, not if we have many features that we want (need) to use
  - ✗ "Lots" of data is relative; n & d large $\neq$ "easy to learn"

Key Questions, Reprise

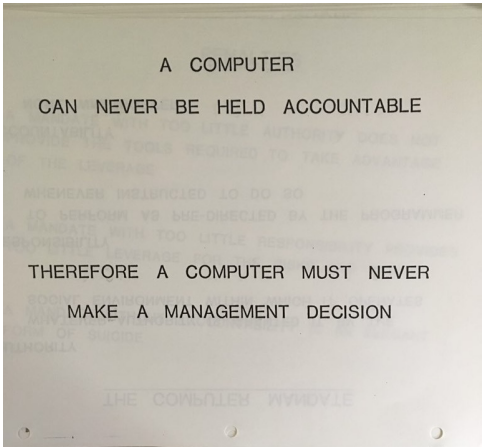
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- Often, we impose many assumptions on our models and data: *Why*?
  - ✗ **Structure matters**: without any additional assumptions (priors), there is not much we can hope to do (apart from getting better data...)
  - ? **Deep learning?** Neural networks are a way of being *even more flexible* by creating *more complex* models. If we want some guidance on what we are doing, we need **even more assumptions**...

IBM training manual (1979)





Thank You!

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